

User Behavior Analysis Based on Stacked Autoencoder and Clustering in Complex Power Grid Environment

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Abstract—Analyzing user behavior characteristics in a complex power grid environment is essential for user behavior planning and resource coordination optimization. Traditional user behavior analysis methods based on model-driven and causal analysis have the disadvantages of strong subjectivity and physical models that are difficult to deal with the randomness and uncertainty of user behavior in complex grid environments. In this paper, we use unsupervised learning methods to analyze user behavior in complex power grid environments, and propose user behavior analysis methods based on stacked autoencoder and clustering. We first reduce the complexity of user behavior data by proposing adaptive feature selection algorithm of user behavior based on stacked autoencoder and unsupervised learning (AFS-SAEUL). Finally, we build a user behavior analysis model based on adaptive feature selection and improved clustering (UBA-AFSIC). The model improved the performance of unsupervised classification of user behavior by fusing the adaptive generation strategy of the initial cluster centers. The simulation experiment results on two real electricity datasets and one public electric vehicle charging dataset show that compared with the existing feature selection algorithm and clustering algorithm, the algorithms proposed in this paper have higher feature selection rate and better clustering performance.

Index Terms—Feature selection, stacked autoencoder, unsupervised learning, power consumption behavior analysis, electric vehicle charging behavior analysis.

I. INTRODUCTION

USER behavior in a complex grid environment includes user electricity consumption, electric vehicle charging

and discharging behavior, and so on. By analyzing user behavior trends and constructing user behaviors of different temporal and spatial scales to describe various user behavior characteristics, we can achieve accurate and unsupervised classification of users, and ultimately provide theoretical and technical support for various user behavior planning and demand-side resource collaborative optimization [1], [2].

Power consumption behavior analysis is an important part of user behavior analysis in complex grid environment. By analyzing electricity consumption behavior, we can describe the electricity consumption characteristics of various users, and ultimately improve the efficiency of electricity consumption, and provide help for grid operation planning [3], [4]. At the same time, with the widespread access of distributed energy, electric vehicles and energy storage, the charge and discharge data of electric vehicles has increased exponentially. From the perspective of automatic control, electric vehicles are more convenient to realize automatic driving [5], [6]. Therefore, in-depth analysis of electric vehicle user behavior provides an important basis for the energy consumption of electric vehicles, the precise classification of electric vehicle users, the installation and deployment of charging piles, and the intelligence of electric vehicle autonomous driving. Fig.1 shows a schematic diagram of user behavior analysis in a complex grid environment.

At present, traditional research methods include model-driven, survey research and causal analysis [7]. Among them, model-driven user behavior analysis describes user behavior by constructing physical models. However, in a complex power grid, user behavior has strong randomness and uncertainty (for example, electricity users will increase the random fluctuation of electricity load due to individual reasons, and electric vehicles will increase the uncertainty of charging and discharging behavior due to environmental uncertainty, etc.) make it difficult to describe user behavior through accurate physical models. User behavior analysis based on survey research and causal analysis is highly subjective, and the accuracy of user classification is not enough, and it is difficult to find the mechanism behind abnormal user behavior. At the same time, these methods do not take into account the high-dimensional characteristics of user behavior data. Therefore, it is very important to construct an accurate user behavior analysis model that tolerates random fluctuations.

To address the above problems, this paper proposes user behavior analysis based on stacked autoencoder and improved clustering in complex power grid environment (UBA-SAEIC).

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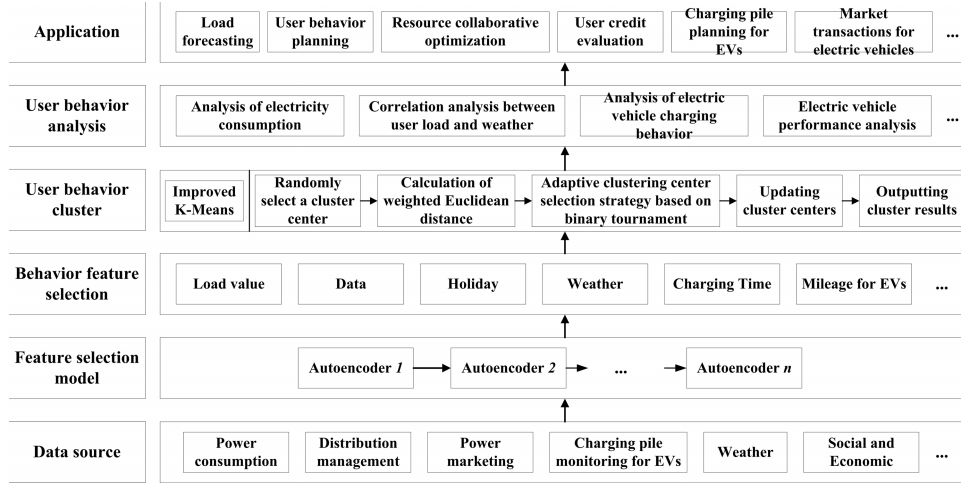


Fig. 1. Schematic diagram of user behavior analysis in complex grid environment.

Regarding electricity consumption behavior data and electric vehicle charging data, on the basis of feature selection through stacked autoencoders, the improved K-Means is used to achieve unsupervised classification of electricity consumption behavior and electric vehicle charging behavior.

In general, the contributions of this work are shown as follows.

- We propose an adaptive feature selection algorithm of user behavior based on stacked autoencoder and unsupervised learning (AFS-SAEUL). By extracting the hidden layer features of the stacked autoencoder, the adaptive selection of features is realized based on K-Means clustering analysis.
- We build a user behavior analysis based on adaptive feature selection and improved clustering (UBA-AFSIC). On the basis of adaptive feature selection, the initial clustering center adaptive generation strategy is integrated, which greatly improves the performance of the K-Means.
- We conduct experiments on two real load datasets and one open dataset for electric vehicles with two tasks of feature selection and user behavior clustering. Experimental results demonstrate the effectiveness of the proposed algorithms in this paper.

The rest of this paper is organized as follows. Section 2 discusses the related work, and methods and models of user behavior analysis are proposed in Section 3. Experimental results and analyses are shown in Section 4. Finally, Section 5 concludes the whole paper.

II. RELATED WORK

In this section, we briefly summarize studies about dimension reduction and clustering for power load data. We then discuss behavior analysis for power consumption and electric vehicle that are closely related to our work.

A. Dimension Reduction and Clustering for Power Load Data

The main purpose of power load data clustering is to obtain the classification of different users and extract the typical

power consumption behavior patterns of users. The improved clustering algorithm is often used in the direct clustering method. McLoughlin *et al.* [8] proposed three of the most widely used unsupervised clustering methods which includes k-means, k-medoid and self-organizing mapping (SOM) for user load mode clustering. By proving the equivalence relation between the number of clusters of load curves and the larger characteristic roots, Ji *et al.* [9] proposed a clustering method based on the analysis of matrix characteristic roots. In addition to common dimensionality reduction methods, wavelet transform is also commonly used in dimensionality reduction of high-dimensional load curves. Luo *et al.* [10] minimized the 96-dimensional features into 8-dimensional ones, and then used DBSCAN clustering algorithm to get load curve classification, which greatly improves the accuracy of load prediction by using the grouped data. In [11], the discrete wavelet transform was used to extract some spectral features and the fuzzy c-means was implemented as the clustering algorithm. Jiang *et al.* [12] obtained the approximate signal and detail signal by multi-stage one-dimensional wavelet transform. Mets *et al.* [13] used fast wavelet transform to reduce the dimension, obtained the typical daily load curve of a single user through the daily load curve, and finally extracted the user load mode through the quadric clustering of the typical load curve of each user. At the same time, some researchers also proposed some new feature construction methods for load curve clustering. Kong *et al.* [14] presented the EMD based CNN-LSTM approach to extract multimodal spatial-temporal features from input data, which significantly improved the accuracy. Al *et al.* [15] put forward a new construction method to extract characteristics, namely calibrated the features on the MC shape and subgroup discovery on the profile error for clustering load curve. Feng *et al.* [16] used the entropy to characterize the longitudinal volatility of customers' power consumption behaviors. Wang *et al.* [3] used SAX and time-based Markov model to model the dynamic characteristics of users' electricity consumption, and then used CFSFDP efficient clustering algorithm to divide users into different categories.

However, due to the high dimension and temporal nature of users' electricity data, the existing researches pay more attention to either the improvement of clustering algorithm or the feature extraction problem, and there are few targeted curve clustering methods that take into account both problems.

B. Behavior Analysis for Electricity Consumption

The analysis of users' electricity consumption behavior is based on accurately obtaining the typical electricity consumption curve of users. Therefore, the existing researches mainly focus on improving the clustering algorithm and constructing load features on the basis of electricity consumption characteristics of each user to obtain the user's electricity consumption pattern. Then carry out analysis on the basis of each typical pattern. Zhao *et al.* [17] tried to cluster the load data after PCA dimension reduction to obtain thirteen different types of load patterns, and then analyzed the electricity characteristics of the 13 load patterns in detail. In [18], by using the upper approximation set and the lower approximation set of the rough set to improve the clustering center formula of K-means, a method of electricity behavior analysis based on fuzzy clustering of the rough set was proposed. In [19], the similar users located in fix area were clustered in accordance with the electricity consumption characteristics and the outlier electricity consumption for the similar users were obtained with density clustering. Lu *et al.* [20] extracted a pattern of user behavior from temperature sensitivity, electricity price sensitivity and power consumption stationarity. Aiming at peak shifting and averting of user side, Wang *et al.* [21] firstly made the classification rules of electricity consumption behavior according to the time of peak consumption, and then used the a hybrid model of deep belief network and extreme learning machine (DBN-ELM) to classify and analyze the peak user group and non-peak user group.

The existing power consumption behavior analysis methods mainly focus on the decomposition of user load to obtain the typical power consumption behavior curve from multiple angles and realize the classification of the power consumption behavior of users. But lack the in-depth analysis of the characteristics of the power consumption behavior of users and the exploration of the factors affecting the power consumption behavior of users.

C. Behavior Analysis for Electric Vehicle

Behavior analysis for electric vehicles is an important branch of user behavior analysis in complex grid environment. At present, researchers mainly use electric vehicle behavior analysis to provide references for electric vehicle track planning, abnormal motion behavior detection, and the design and deployment of charging pile locations. Lv *et al.* [22] explored a route guidance strategy of electric vehicle by using improved FCM clustering algorithm in big data analysis technology. Campo *et al.* [23] detected abnormal behavior of real vehicle by using Gaussian process regression. Fazeli *et al.* [24] proposed an improved two-stage stochastic programming model to decide the optimal layout of electric vehicle supply equipment by analyzing drivers behaviors.

Zhang *et al.* [25] predicted energy consumption of electric vehicles based on machine learning by using real-world driving data in Beijing city to support for the development of electric vehicles equipments. Xiong *et al.* [26] used unsupervised clustering algorithm to predict EV driver behaviors in order to reduce the impact of uncertain environments on EV charging or discharging scheduling. In [27]–[33], researchers focused on charging management between multiple EVs by analyzing all kinds of behaviors.

However, existing electric vehicle behavior analysis focuses on charging and discharging management, electric vehicle abnormal behavior detection, and trajectory analysis. It does not take into consideration the segmentation of electric vehicle user behavior, and cannot provide targeted strategies for various user behaviors in the later period.

III. METHODS AND MODELS

A. Grey Relational Analysis

The basic idea of gray correlation analysis is to determine whether the relationship is close by determining the geometric similarity of the reference sequence and several comparison sequences, which reflects the degree of correlation between the curves [34]. To better analyze the correlation relationship between user behavior data and the characteristic data affecting behavior in a complex power grid environment, this paper uses gray correlation analysis to assist in analyzing the inherent relationship between the user behavior curve and the characteristic curve formed after unsupervised clustering.

Suppose that the reference sequences of user behavior are $Y = y(k), k = \{1, 2, \dots, n\}$, and the comparison sequences that affect user behavior are $X_i = x_i(k), k = \{1, 2, \dots, n\}, i = \{1, 2, \dots, m\}$.

Since the dimensions of user load and the charging time of electric vehicles are different from the dimensions of the meteorological factors that affect user load and the mileage of electric vehicles after being fully charged, we cannot directly use grey correlation analysis for calculation. To better calculate the degree of correlation between the reference series and the comparison series, we need to perform dimensionless processing on the above reference sequences and the comparison sequences. The processed reference sequences and the comparison sequences are $Y' = y'(k), k = \{1, 2, \dots, n\}$, $X'_i = x'_i(k), k = \{1, 2, \dots, n\}, i = \{1, 2, \dots, m\}$, respectively.

Definition 1 ([34]): Let $Y' = y'(k), k = \{1, 2, \dots, n\}$ be the dimensionless reference sequences, and $X'_i = x'_i(k), k = \{1, 2, \dots, n\}, i = \{1, 2, \dots, m\}$ be the dimensionless comparison sequences. Let $\varepsilon_i(k) = |y'(k) - x'_i(k)|$, we have $\mu_i(k) = \frac{\min_k \min_i \varepsilon_i(k) + \lambda \max_k \max_i \varepsilon_i(k)}{\varepsilon_i(k) + \lambda \max_k \max_i \varepsilon_i(k)}$. Then $\mu_i(k)$ is called the correlation coefficient between the reference sequences and the comparison sequences.

Where $\lambda \in (0, \infty)$ is the resolution coefficient, the smaller the λ , the stronger the resolution. In this paper, $\lambda = 0.5$.

Definition 2: Let $\mu_i(k)$ be the correlation coefficient between the i -th data in the comparison sequence and the k -th data in the reference sequence. Then $r_i = \frac{1}{n} \sum_{k=1}^n w_k \mu_i(k)$

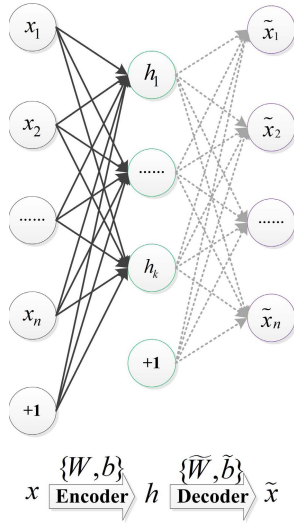


Fig. 2. The structure of AE model.

is called the correlation degree between the i -th data in the comparison sequence and the reference sequence.

Where w_k represents the weight of the k -th data in the reference sequence.

B. AFS-SAEUL

There are many features that affect user behavior in the complex grid environment. If the user behavior is directly clustered and analyzed, the results are complicated, and it is difficult to find user behavior patterns and patterns. To reduce the number of features that affect user behavior, this paper proposes an adaptive feature selection algorithm of user behavior based on stacked autoencoder and unsupervised learning (AFS-SAEUL).

1) *Stack AutoEncoder*: Autoencoder (AE) is a neural network that first compresses the input into a latent spatial representation, and then reconstructs the output through this representation [35], [36]. It mainly includes two processes: the encoding process from the input to the hidden layer and the decoding process from the hidden layer to the output. The purpose of the AE is to reconstruct the original input as much as possible. The structure of the AE model is shown in Fig.2.

The encoder uses a mapping function f to project x from the input layer to the hidden layer $h = [h_{(1)}, h_{(2)}, \dots, h_{(m)}]$, where m is the dimension of the hidden layer variable vector. The mapping function is $f(x) = s_f(Wx+b)$, W is a weight matrix of $n \times m$ dimension, b is deviation vector. The decoder maps the h represented by the hidden layer to the output layer $\tilde{x}$ through the mapping function $\tilde{f}$. The mapping function is $\tilde{f}(h) = s_g(\tilde{W}h+\tilde{b})$, $\tilde{W}$ is a weight matrix of $n \times m$ dimension, $\tilde{b}$ is a deviation vector.

The ultimate goal of the AE is to make $|\tilde{x} - x|$ close to 0 infinitely by setting the number of neurons in the hidden layer. To obtain the AE model parameters, the mean square reconstruction error is calculated by Equation (1) to minimize the loss function, and the parameters of AE are updated by the gradient descent method. Finally, under the condition that $|\tilde{x} - x|$ is infinitely close to 0, the current autoencoder



Fig. 3. The structure of SAE.

parameters are the trained autoencoder parameters.

$$J(W, \tilde{W}, b, \tilde{b}) = \sum_{i=1}^N ||\tilde{x}_i - x_i||^2 / 2N \quad (1)$$

Stacked Autoencoder (SAE) is formed by connecting multiple autoencoders layer by layer, and the i -th encoder maps the original data to the i -th hidden layer. We then use the i -th hidden layer data as the input of the $i+1$ -th autoencoder, and then train the $i+1$ -th hidden layer to obtain the parameters $\{W_{i+1}, b_{i+1}\}$. Finally, the hidden layer data is used as the input of the next AE. The structure of the SAE is shown in Fig.3.

2) Description of Feature Selection for User Behaviors:

Both autoencoder and principal component analysis are typical unsupervised learning methods and are often used in feature selection [37], [38]. The AE realizes that the error between output and input is as close to 0 as possible by minimizing the loss function for multiple iterations. The hidden layer data of the trained autoencoder is considered to be the feature data after dimensionality reduction. However, in the SAE, the number of hidden layers cannot be determined theoretically, and can only be randomly selected within a certain range. We can use unsupervised or supervised learning to verify whether the current number of hidden layers is optimal, thereby determining the feature dimension. Therefore, to better adaptively select features for user behavior data, this paper proposes user behavior adaptive feature selection based on K-Means clustering algorithm. The framework is shown in Fig.4.

Let $B = \{B_1, B_2, \dots, B_n\}$ be the user behavior dataset. After calculation, the number of hidden layers in the current stack autoencoder is $H \in [p, q]$. Where p and q are the upper and lower bounds of the number of hidden layers in the stack autoencoder, and satisfy $p, q < n$.

The number of hidden layers randomly set in the stacked autoencoder is clustered by the improved K-Means. And by comparing the Davies Bouldin Index (DBI) of the improved K-Means under different number of hidden layers, the optimal number of hidden layers can be adaptively selected.

Suppose that the user behavior dataset $B = \{B_1, B_2, \dots, B_n\}$ has k clusters $C = \{C_1, C_2, \dots, C_k\}$, the dispersion degree $avg(C_i)$ of the user behavior dataset B can be solved by the following equation

$$avg(C_i) = \frac{1}{N_i} \sum_{j=1}^{N_j} |B_j - C_i|. \quad (2)$$

where N_i represents the number of user behavior data in the i -th cluster, U_j represents the j -th data point in the i -th cluster, and C_i represents the center of the i -th cluster.

The distance between any two clusters (C_i, C_j) in k clusters $C = \{C_1, C_2, \dots, C_k\}$ is obtained by Equation (3)

$$d_{cen}(C_i, C_j) = \sum_{m=1}^N |c_{mi} - c_{mj}|. \quad (3)$$

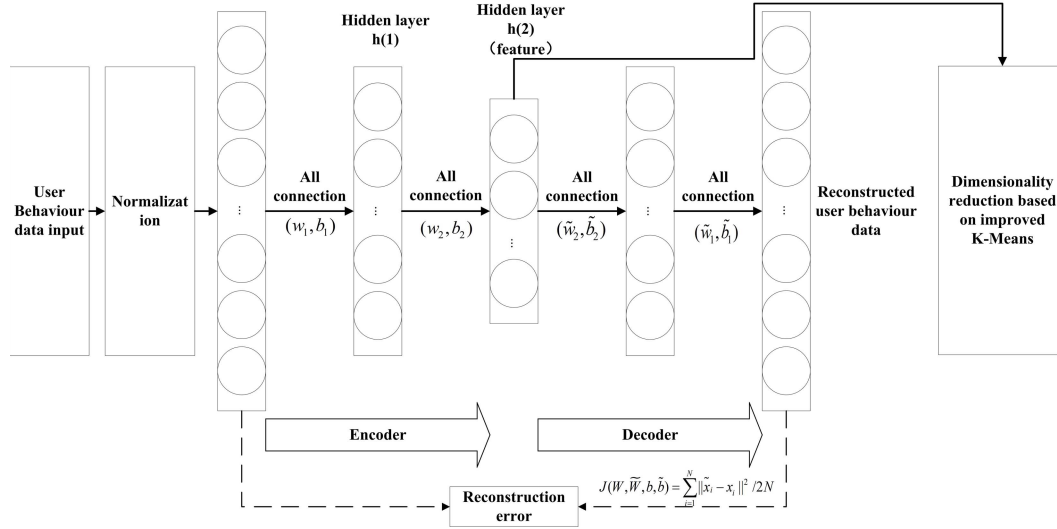


Fig. 4. Feature selection framework of user behavior based on K-Means and stacked autoencoder.

where N represents the dimension of any cluster center, and c_{mi} represents the m -th attribute value of the i -th type center point.

Definition 3: Let $C = \{C_1, C_2, \dots, C_k\}$ be k clusters, $avg(C_i)$ represents the average distance from all sample data in the i -th cluster to the cluster center, $d_{cen}(C_i, C_j)$ represents the distance between the i -th cluster center and the j -th cluster center. Then $\frac{1}{k} \sum_{i=1}^k \max_{j \neq i} \left(\frac{avg(C_i) + avg(C_j)}{d_{cen}(C_i, C_j)} \right)$ is called the Davies Bouldin Index (DBI) of k clusters.

The smaller the DBI, the better the cluster effect.

AFS-SAEUL is shown as follows.

C. UBA-AFSIC

The essence of user behavior analysis in complex power grids is to use unsupervised learning algorithms to cluster unlabeled user behavior data. Then we analyze user characteristics from the clustering results, classify user types, and provide references and basis for later user planning.

K-Means is a typical unsupervised learning algorithm [39]. Based on the K-Means, user behavior data under complex power grids can be effectively clustered, and it is easy to find the difference between multiple types of user characteristic curves to analyze user behavior, and then perform unsupervised classification according to user behavior. However, the initial clustering center generation of the traditional K-Means algorithm has strong randomness, which causes the algorithm to converge slowly and easily fall into local convergence. Therefore, this paper proposes a user behavior analysis algorithm based on improved K-Means.

Since we cannot guarantee the consistency of user behavior data distribution in complex grid environment, this paper uses weighted Euclidean distance to calculate the distance between sample data. Let $U = \{U_1, U_2, \dots, U_n\}$ be the user behavior data vector, $M = \{M_1, M_2, \dots, M_n\}$ be the corresponding mean vector, and $S = \{S_1, S_2, \dots, S_n\}$ be the standard deviation vector. Then the normalized vector of user behavior

data shown in Equation (4) is formed

$$U^* = \{U_1^*, U_2^*, \dots, U_n^*\}. \quad (4)$$

where $U_i^* = \frac{U_i - M_i}{S_i}$ represents the normalized vector of the i -th row of user behavior data.

For any two normalized vector $U_i^*, U_j^* (i \neq j)$ of user behavior data, the weighted Euclidean distance is shown in Equation (5).

$$dist(U_i^*, U_j^*) = \sqrt{\sum_{k=1}^n \left(\frac{u_{ik}^* - u_{jk}^*}{S_k} \right)^2} \quad (5)$$

Let $dist(U_i^*, U_j^*) = d_{ij}$.

On the basis of calculating the weighted Euclidean distance, to better improve the randomness of the initial cluster center selection of the traditional K-Means, we use the binary tournament selection strategy in the evolutionary algorithm to select the cluster center of the K-Means.

Firstly, an initial cluster center $U_i^*, i \in \{1, 2, \dots, n\}$ is randomly selected from the user behavior data normalized vector $U^* = \{U_1^*, U_2^*, \dots, U_n^*\}$, and then the weighted Euclidean distance set $d = \{d_{ij}\}, i, j = \{1, 2, \dots, n\}$ between the remaining user behavior data normalized vector $U_j^*, j = \{1, 2, \dots, n\}$ and the initial cluster center $U_i^*, i \neq j$ is calculated by Equation (5).

Secondly, according to the probability p , two distances d_{ij}, d_{ik} are randomly selected from d . If $d_{ij} < d_{ik}$, the k -th user behavior data is selected as the second cluster center. The above operation is repeated until all initial cluster center vectors $C = \{C_1, C_2, \dots, C_K\}$ are selected.

Then, for each data vector $U_i^*, i = \{1, 2, \dots, n\}$ in $U^* = \{U_1^*, U_2^*, \dots, U_n^*\}$, we calculate the weighted Euclidean distance d_{ip}^* between each cluster center in U_i^* and $C_p, p = \{1, 2, \dots, K\}$ to form an updated set $d = \{d_{ip}^*\}$ of weighted Euclidean distances. Then U_i^* is classified into the cluster set corresponding to the minimum $\min\{d_{ip}^*\}$ in $\{d_{ip}^*\}$.

Then, according to all user behavior data in the current cluster center vector, we calculate the cluster center again

Algorithm 1 : AFS-SAEUL

Input: $B = \{B_1, B_2, \dots, B_n\}$, N , $Relu$;
Output: $m(\text{dimension of sampled data after feature selection})$;
/*Input data into stack autoencoder */
1. $SAE(W, \tilde{W}, b, \tilde{b}) \leftarrow \text{Input}(B, N, Relu)$;
/* Consider p as the number of clusters in the K-Means */
2. **for** $p \leftarrow 1$ to $N - 1$ **do**
3. $C \leftarrow K - \text{Means}(B, p)$;
4. **for** $i \leftarrow 1$ to p **do**
5. $avg(C_i) \leftarrow \frac{1}{N_i} \sum_{j=1}^{N_j} |B_j - C_i|$;
6. **for** $j \neq i, j \in [1, p]$ **do**
/* Calculate the distance between any two clusters */
7. $d_{cen}(C_i, C_j) \leftarrow \sum_{k=1}^N |c_{ki} - c_{kj}|$;
8. **end for**
9. **end for**
/* Calculate the DBI value of the current cluster */
10. $DBI_p \leftarrow \frac{1}{k} \sum_{i=1}^k \max_{j \neq i} \left(\frac{avg(C_i) + avg(C_j)}{d_{cen}(C_i, C_j)} \right)$;
11. **end for**
12. $I \leftarrow DBI_1$;
13. **for** $p \leftarrow 2$ to $N - 1$ **do**
14. **if** $DBI_p < I$ **then**
15. $I \leftarrow DBI_p$;
16. **end if**
17. **end for**
18. $m \leftarrow p$;
19. **return** m ;

according to Equation (6) and update the cluster center vector C .

$$C_k = \frac{1}{|C_k|} \sum_{U_i \in C_k} U_i^* \quad (6)$$

Finally, we calculate the difference ΔJ of the weighted Euclidean distance between U_i^* and the cluster center C_k (as shown in Equation (7)) in the two iterations. If $\Delta J < \varepsilon$, we consider that the algorithm has converged.

$$J = \sum_{k=1}^K \sum_{U_i \in C_k} d(U_i^*, C_k) \quad (7)$$

UBA-AFSIC is shown as follows.

The UBA-AFSIC is mainly composed of feature selection for user behavior data, initial cluster center generation, calculation of weighted Euclidean distance, and cluster center update. The time complexity is mainly $O(n * t * l)$, $O(n^2)$, $O(n^2 + K)$, and $O(n^2)$, respectively, where n represents the size of user behavior data, t represents the time required for the autoencoder to calculate a piece of data, l represents the number of iterations of the autoencoder, and K represents the number of clusters. In actual situations, $n \gg t$ and $n \gg l$, the time complexity of the UBA-AFSIC algorithm is about $O(n^2)$.

Algorithm 2 : UBA-AFSIC

Input: $U = \{U_1, U_2, \dots, U_n\}$, N , $Relu$, K , ε , p , T ;
Output: U_C ;
/*Feature selection of user behavior data*/
1. $m \leftarrow AFS - SAEUL(U, N, Relu)$;
/*Normalization of user behavior data */
2. **for** $i \leftarrow 1$ to n **do**
3. $U_i^* \leftarrow \frac{U_i - M_i}{S_i}$;
4. **end for**
5. $U_i^* \leftarrow \text{RanSeleClusterCenter}(U^*)$;
/*Insert the first randomly selected cluster center into U_C */
6. $U_C \leftarrow \text{Insert}(U_i^*)$;
7. **for** $i \neq j, j \in \{1, 2, \dots, n\}$ **do**
8. $d_{ij} \leftarrow \sqrt{\sum_{k=1}^n \left(\frac{u_{ik}^* - u_{jk}^*}{S_k} \right)^2}$;
9. **end for**
/*Construct the cluster center vector U_C according to the binary tournament strategy*/
10. **while** $|U_C| \neq m$ **do**
11. $d_{ij} \leftarrow \text{RanSelect}(d, p)$;
12. $d_{ik} \leftarrow \text{RanSelect}(d, p)$;
13. **if** $d_{ij} < d_{ik}$ **then**
14. $U_C \leftarrow \text{Insert}(U_k^*)$;
15. **end if**
16. **end while**
17. **for** $t \leftarrow 1$ to T **do**
18. **for every** $U_i^*, i \in \{1, 2, \dots, n\}$ **do**
19. $d \leftarrow \text{dist}(U_i^*, U_C)$;
20. $U_C.\text{insert}(\min(d))$;
21. **end for**
22. Update the cluster center vector U_C according to Equation (6);
23. Calculate the ΔJ of two iterations according to Equation (7);
24. **if** $\Delta J < \varepsilon$ **then**
25. **return** U_C ;
26. **break**;
27. **end if**
28. **end for**

IV. EXPERIMENTS AND ANALYSIS

In this section, we conduct experiments on two real datasets and one public accessible datasets to evaluate the performance of the proposed algorithms in this paper. The experimental setup is first described. Then, experiments are conducted to demonstrate the advantage of the proposed algorithms, respectively.

A. Experimental Setup

1) *Dataset*: The experiments are conducted on two real datasets and one public dataset, which are representative of two types of user behavior: power consumption behavior and electric vehicle charging and discharging behavior. And to better analyze the relationship between various load data and weather data, all load datasets are divided into three sections

TABLE I
STATISTICS OF EXPERIMENTAL DATASETS

	Dataset	Date	#Instance	#Feature
Data1	summer	08/01/2019-09/20/2019	51	96
	winter	12/02/2019-02/20/2020	81	
	spring & autumn	09/21/2019-12/01/2019	119	
	summer	21/02/2020-04/07/2020		
Data2	summer	08/01/2019-09/20/2019	51	96
	winter	12/02/2019-02/20/2020	81	
	spring & autumn	09/21/2019-12/01/2019	119	
	summer	21/02/2020-04/07/2020		
Data3	/	/	98	11

TABLE II
STATISTICS OF CORRESPONDING WEATHER DATASETS

Dataset	#Instance	#Feature	Date
WeatherData1	251	9	08/01/2019-04/07/2020
WeatherData2	251	9	08/01/2019-04/07/2020

according to summer, winter, spring & autumn. The statistics of the three experimental datasets are shown in Table I. And in order to analyze the electricity consumption behavior from the two real load data, we also added the weather data of the corresponding date, as shown in Table II.

Data1 contains 251-day load data from August 1, 2019 to April 7, 2020 in a station area of Nantong Power Supply Company of China. Among them, the load data collection interval is 15 minutes, and each row of data in Data1 contains the load values recorded at 96 sampling points. And statistic of Data2 is the same as Data1¹. The corresponding weather dataset includes nine weather factors including daily minimum temperature, daily maximum temperature, daily average temperature, average relative humidity, average wind speed, daily rainfall, sunshine hours, average surface temperature, and average air pressure. The only difference between Data1 and Data2 is that they collect load data from different station area of Nantong Power Supply Company of China.

Data3 is a dataset about electric vehicles², with a total of 98 data, each of which includes 13 features. In this paper, we performed standard discretization on the four features which include PowerTrain, PlugType, BodyStyle, and Segment in Data3, and removed the two features such as Brand and Model. Therefore, in the actual experiment, the feature number of Data3 is 11.

2) *Evaluation Indexes*: In this paper, we use feature selection ratio, Davies Bouldin Index (DBI) and Silhouette Coefficient (SC) to judge the performance of proposed algorithm. The definitions of evaluation indexes are shown as follows.

Definition 4: Let $X = \{X_1, \dots, X_n\}$ be the user behavior dataset, $F = \{F_1, \dots, F_m\}$ be the feature set, $X' = \{X'_1, \dots, X'_n\}$ be the user behavior dataset after feature selection, and $F' = \{F'_1, \dots, F'_t\}$ be the feature set after feature selection. Where $t < m$. Then $f_r = \frac{n-t}{n}$ is called the feature selection ratio.

¹<https://github.com/Namanana-CC/Load>

²<https://www.kaggle.com/geoffnel/evs-one-electric-vehicle-dataset>

At the same time, in addition to the Davies Bouldin Index (DBI) described in *Definition 3*, the performance measurement of the clustering algorithm also uses the Silhouette Coefficient (SC) as the evaluation index.

Definition 5: Let $C = \{C_1, C_2, \dots, C_k\}$ be k clusters, a_i be the average distance from sample i to other samples in the same cluster, and b_{ij} be the average distance from sample i to all samples in other clusters C_j . Denote $b_i = \min\{b_{i1}, b_{i2}, \dots, b_{ik}\}$. Then $s_i = \frac{b_i - a_i}{\max\{a_i, b_i\}}$ is called the Silhouette Coefficient of sample i .

The larger the SC, the better the clustering effect.

3) *Comparative Methods*: In this experiment, we compare algorithms from two aspects: feature selection and user behavior cluster analysis. For feature selection, we compare AFS-SAEUL with Principal Component Analysis (PCA) [40] and Sparse Random Projection (SRP) [41]. And for user behavior cluster analysis, we compare UBA-AFSIC with Agglomerative Clustering (AC) [42], Gaussian Mixed Model (GMM) [43], Spectral Clustering (SC) [44], Fuzzy C-Mean (FCM) [45] and Minibatch K-Means (MKmeans) [46]. All Comparison algorithms are shown in Table III.

4) *Parameter Settings*: In this experiment, we listed the experimental parameters of PCA, SRP, AFS-SAEUL, AC, GMM, SC, FCM, MKmeans, Kmeans and UBA-AFSIC, respectively. The parameter settings of all algorithms are shown in Table IV and Table V.

B. Experimental Analysis of Feature Selection

The feature selection of user behavior not only reduces the complexity of user behavior analysis and modeling, but also improves the visualization of user behavior analysis and modeling. For the experimental datasets in Table I, Fig.5 shows the changing trend of the feature selection based on AFS-SAEUL under the condition of different number of second hidden layers. And for the experimental datasets in Table I, under the optimal feature dimension conditions, we compare AFS-SAEUL with PCA and SRP. The experimental results are shown in Fig.6.

From Fig.5(a), (b) and (c), we find that when the number of second hidden layers of SAE is 7, 9, and 3, the DBI value obtained by AFS-SAEUL is the smallest, which are 0.8858, 0.7156 and 0.4076, respectively. According to Definition 3, we know that the smaller the DBI value, the better the feature selection effect of AFS-SAEUL. And from Fig.6, we can also see that the feature selection ratio of AFS-SAEUL is superior to PCA and SRP for all experimental datasets in Table I.

From Fig.5, we know that the optimal results of AFS-SAEUL for Data1, Data2 and Data3 are 7, 9, and 3, respectively. Therefore, we set 7, 9, and 3 as the number of clusters in the improved K-Means, and use DBI and SC to evaluate the pros and cons of feature selection algorithms. From Fig.7, we can see that for the three experimental datasets in Table I, compared with PCA and SRP, AFS-SAEUL has the smallest DBI and the largest SC. This indicates that clustering the features extracted by the stack autoencoder can ensure a better clustering effect. Specifically, for Data1, Data2, and Data3, compared with PCA, the DBI of AFS-SAEUL

TABLE III
COMPARISON METHODS

Feature Selection Algorithm	Description	Cluster Algorithm	Description
PCA	Principal component analysis (PCA) is an unsupervised linear data conversion technique widely used in different fields. It finds the direction of maximum variance in high-dimensional data by calculating the eigenvalues of the covariance matrix and the corresponding eigenvectors, and maps the data to a new subspace with a dimension not greater than the original data.	AC	Agglomerative Clustering (AC) can divide the dataset at different levels to form a tree-like cluster structure. The algorithm treats each sample as one class, and then clusters them according to the similarity between clusters until all samples are clustered into one class.
SRP	This method uses a sparse random matrix and reduces the dimensionality by projecting the original input space.	GMM	The algorithm is formed by linear superposition of multiple Gaussian models, and the probability of each class is obtained.
		SC	SC is a clustering method based on graph theory, which clusters the eigenvectors of the Laplacian matrix of sample data to achieve the purpose of clustering sample data.
		FCM	FCM obtains the membership degree of each sample point to all cluster centers by optimizing the objective function, thereby determining the class of sample points to achieve the purpose of automatically classifying sample data.
		MKmeans	It uses a small data subset (the data set used for each training is a randomly selected data subset) to reduce the calculation time, while trying to optimize the objective function.

TABLE IV
EXPERIMENTAL PARAMETERS FOR FEATURE SELECTION ALGORITHMS

Parameter	PCA	SRP	AFS-SAEUL
Input	251*96(Data1,2) 98*11(Data3)	251*96(Data1,2) 98*11(Data3)	251*96(Data1,2) 98*11(Data3)
First hidden layer	/	/	48(Data1,2) 11(Data3)
Second hidden layer	/	/	7(Data1)/9(Data2) 3(Data3)
whiten	False	/	/
eps	/	0.1	/
n_components	7	7	/

decreased by about 27.29%, 29.73%, and 33.24%, respectively; compared with SRP, the DBI of AFS-SAEUL decreased by about 35.22%, 47.67%, and 46.72%, respectively. At the same time, for the above experimental datasets, compared with PCA, the SC of AFS-SAEUL was increased by approximately 15.65%, 24.81%, and 20.7%; compared with SRP, the SC of AFS-SAEUL was increased by about 39.97%, 51.88%, and 35.88%, respectively.

In Fig.8, for Data1, Data2 and Data3, we use the optimal feature dimensions of AFS-SAEUL, PCA and SRP as the number of clusters in the improved K-Means, and use the DBI and SC to evaluate the performance of algorithms. Under the condition that the optimal feature dimensions are obtained based on different feature selection algorithms, AFS-SAEUL has the smallest DBI and the largest SC.

TABLE V
EXPERIMENTAL PARAMETERS FOR USER BEHAVIOUR ANALYSIS ALGORITHMS

Parameter	AC	GMM	SC	FCM	MKmeans	Kmeans	UBA-AFSIC
Input	251*7(Data1)/251*9(Data2)/98*3(Data3) 7(Data1)/6(Data2)/3(Data3)						
n_clusters	/	100	/	1000	30	10	30
max_iter	/	0.001			/		0.0001
tol							None
random_state				/			
n_components	/	3			/		
gamma		/	0.1			/	
c		/		3		/	
m		/		2		/	
error		/		0.005		/	
batch_size		/			100		/
reassignment_ratio		/			0.01		/
max_no_improvement		/			10		/

C. Experimental Analysis of Power Consumption Behavior

1) *Comparison of Cluster Performance Evaluation:* The analysis of power consumption behavior is essentially to analyze the power consumption characteristic curve by using a clustering algorithm to obtain different power consumption behaviors, and analyze the causes and influencing factors of these behaviors. For Data1 and Data2, on the basis of AFS-SAEUL, Fig.9 compares the performance of different clustering algorithms. From Fig.9, we can see that for Data1 and Data2, compared with other clustering algorithms,

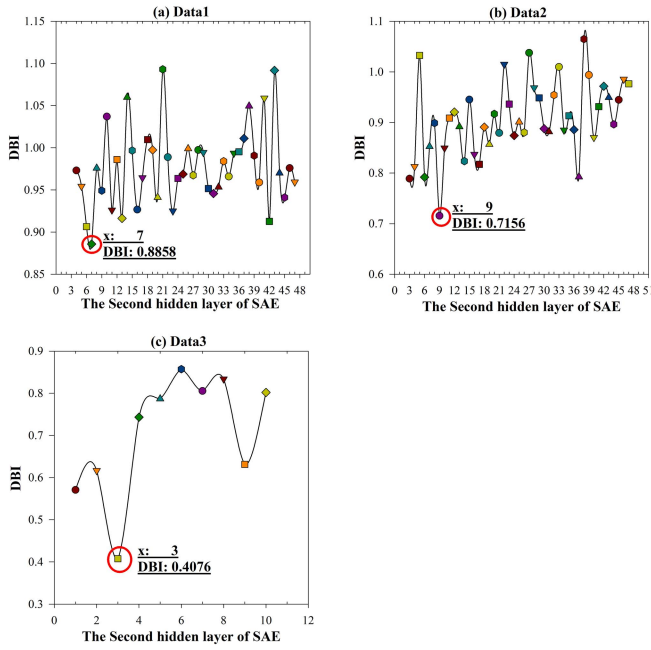


Fig. 5. The changing trend of feature selection of AFS-SAEUL under different number of second hidden layers.

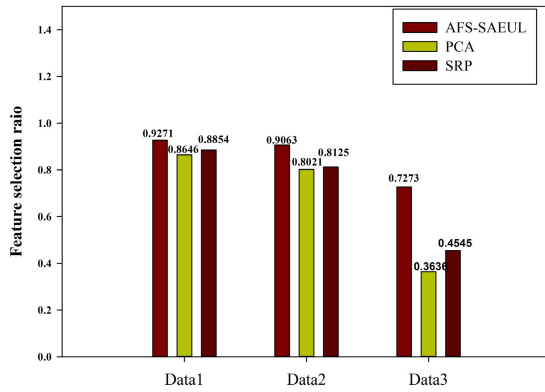


Fig. 6. Comparison of feature selection ratio between AFS-SAEUL, PCA and SRP.

the UBA-AFSIC proposed in this paper has the best evaluation index of clustering performance. Specifically, compared with AC, GMM, SC, Kmeans, FCM, and MKmeans, for Data1 and Data2, the DBI of UBA-AFSIC is the smallest with values of 0.8858 and 0.7156, respectively, and the SC of UBA-AFSIC is the largest with values of 0.4401 and 0.4289, respectively.

2) *Correlation Analysis Between Weather Features and Load Features:* To better analyze the impact of weather on power consumption behavior, for Data1 and Data2 in Table I and WeatherData1 and WeatherData2 in Table II, we analyze the main weather characteristics that affect load consumption in different seasons, and the impact of weather characteristics on user load behavior by calculating the gray correlation between weather characteristics and the three types of load characteristics. Weather characteristics include day minimum temperature, daily maximum temperature, daily average temperature, average relative humidity, average wind speed, daily rainfall, sunshine hours, average surface temperature, and

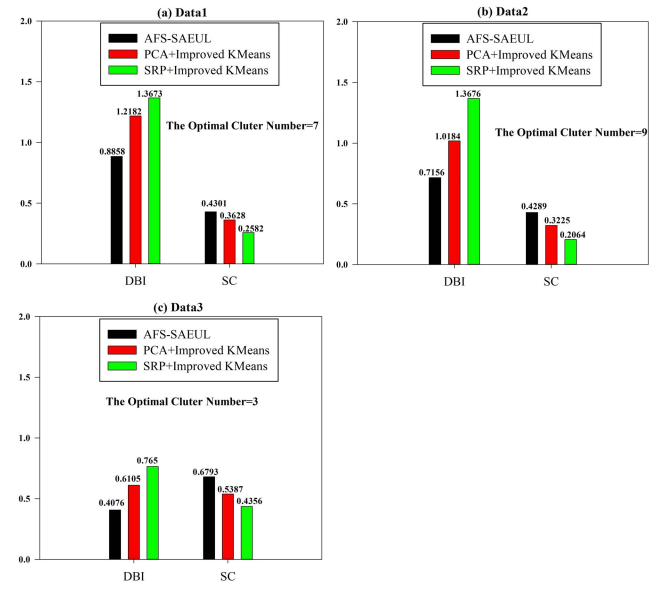


Fig. 7. Comparison of cluster performance between the different algorithms under the optimal number of feature selection for experimental datasets in Table I.

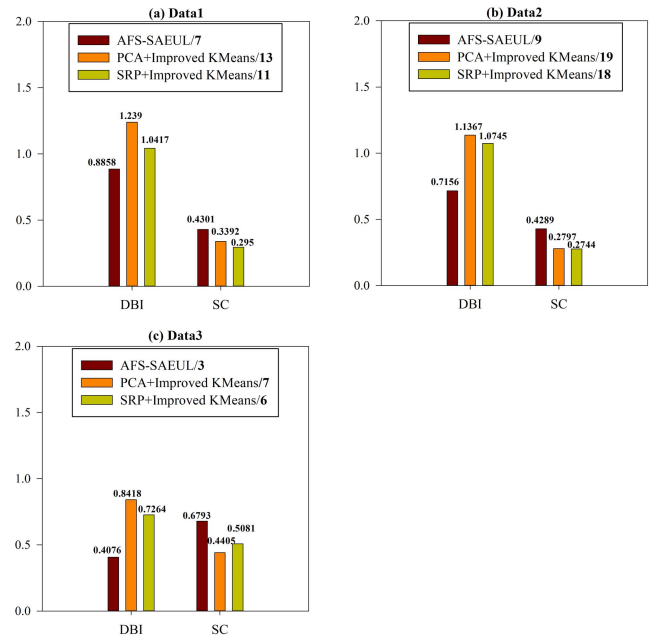


Fig. 8. Comparison of experimental results between AFS-SAEUL, PCA and SRP under the different optimal cluster number for experimental datasets in Table I.

average air pressure. And load characteristics include daily average load, daily peak valley difference and daily maximum load. The results of the correlation degree are shown in Table VI, VII and VIII, respectively.

From the gray correlation analysis, we can see that for summer load, temperature is the main influencing factor of summer load, followed by wind speed. For winter load, the main influencing factors are relative humidity, daily rainfall, and average air pressure. The temperature difference between morning and evening is large in winter, so the difference between daily peak and valley, daily maximum load are also

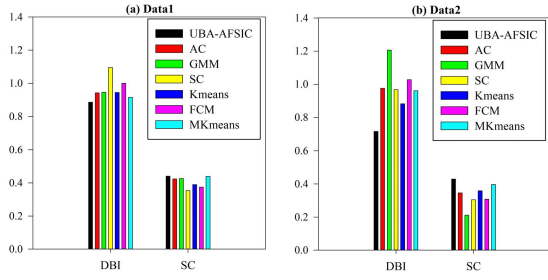


Fig. 9. Comparison of cluster performance evaluation among the different clustering algorithms for data1 and data2 based on feature selection.

TABLE VI

ANALYSIS OF INFLUENCING FACTORS FOR ALL DATASETS IN SUMMER

Dataset	Influencing factors	Load value		
		Average daily load	Peak valley difference	Daily maximum load
Data1	$D_{\min t}$	0.83	0.8	0.8
	$D_{\max t}$	0.83	0.8	0.8
	D_{meant}	0.88	0.83	0.86
	A_{rh}	0.67	0.67	0.65
	A_{wv}	0.74	0.75	0.7
	D_{ra}	0.68	0.69	0.65
	S_d	0.69	0.68	0.69
	M_{st}	0.85	0.82	0.82
	A_{ap}	0.64	0.65	0.61
	$D_{\min t}$	0.83	0.82	0.83
Data2	$D_{\max t}$	0.85	0.86	0.85
	D_{meant}	0.90	0.88	0.89
	A_{rh}	0.68	0.68	0.68
	A_{wv}	0.75	0.73	0.74
	D_{ra}	0.69	0.67	0.68
	S_d	0.70	0.69	0.69
	M_{st}	0.87	0.86	0.86
	A_{ap}	0.66	0.66	0.65

TABLE VII

ANALYSIS OF INFLUENCING FACTORS FOR ALL DATASETS IN WINTER

Dataset	Influencing factors	Load value		
		Average daily load	Peak valley difference	Daily maximum load
Data1	$D_{\min t}$	0.68	0.71	0.70
	$D_{\max t}$	0.67	0.70	0.68
	D_{meant}	0.69	0.72	0.71
	A_{rh}	0.71	0.75	0.73
	A_{wv}	0.66	0.70	0.69
	D_{ra}	0.73	0.75	0.74
	S_d	0.59	0.57	0.59
	M_{st}	0.68	0.72	0.70
	A_{ap}	0.74	0.72	0.75
	$D_{\min t}$	0.71	0.69	0.69
Data2	$D_{\max t}$	0.68	0.69	0.69
	D_{meant}	0.72	0.71	0.71
	A_{rh}	0.73	0.72	0.72
	A_{wv}	0.67	0.69	0.68
	D_{ra}	0.75	0.74	0.75
	S_d	0.59	0.61	0.60
	M_{st}	0.71	0.70	0.70
	A_{ap}	0.72	0.75	0.74

affected by temperature. In spring and autumn, the weather is cool, and user load is not greatly affected by temperature, but is mainly affected by relative humidity, average wind speed, daily rainfall, and average air pressure.

TABLE VIII
ANALYSIS OF INFLUENCING FACTORS FOR ALL DATASETS
IN SPRING & AUTUMN

Dataset	Influencing factors	Load value		
		Average daily load	Peak valley difference	Daily maximum load
Data1	$D_{\min t}$	0.67	0.67	0.66
	$D_{\max t}$	0.66	0.66	0.65
	D_{meant}	0.65	0.66	0.65
	A_{rh}	0.75	0.75	0.75
	A_{wv}	0.77	0.77	0.76
	D_{ra}	0.82	0.82	0.81
	S_d	0.66	0.65	0.66
	M_{st}	0.66	0.66	0.65
	A_{ap}	0.80	0.80	0.80
	$D_{\min t}$	0.74	0.70	0.71
Data2	$D_{\max t}$	0.72	0.69	0.70
	D_{meant}	0.72	0.68	0.70
	A_{rh}	0.81	0.76	0.79
	A_{wv}	0.75	0.72	0.74
	D_{ra}	0.84	0.81	0.82
	S_d	0.64	0.63	0.64
	M_{st}	0.73	0.69	0.70
	A_{ap}	0.79	0.76	0.78

TABLE IX

THE NUMBER OF DAYS INCLUDED IN EACH TYPE OF LOAD CURVE AND THE PROBABILITY OF OCCURRENCE IN DIFFERENT SEASONS FOR DATA1

Type of load curve	Number of days	Number of days /probability of occurrence		
		Spring&Fall	Summer	Winter
Cluster 1	14	0/0	0/0	14/1
Cluster 2	51	32/0.627	11/0.216	8/0.157
Cluster 3	30	2/0.067	12/0.4	16/0.533
Cluster 4	85	77/0.906	8/0.094	0/0
Cluster 5	40	8/0.2	8/0.2	24/0.6
Cluster 6	12	0/0	12/1	0/0
Cluster 7	19	0/0	0/0	19/1

3) *Power Consumption Behavior Analysis*: In this subsection, to analyze power consuming behavior for Data1 and Data2, we use UBA-AFSIC to realize the cluster of daily load and insert the related label. The clustering results are shown in Fig.10 and Fig.11, respectively. From Fig.10 and Fig.11, we can see that the user load data in Data1 and Data2 are clustered into 7 and 6 classifications, respectively. The result shows that the user load in Data1 and Data2 contains 7 and 6 behavior features. After clustering, we insert corresponding classification labels into Data1 and Data2 to facilitate targeted analysis in the later stage.

Based on Fig.10 and 11, we construct a typical daily load curve by calculating the average daily load value of each type of load curve in Data1 and Data2, respectively. The results are shown in Fig.12 and 13. Table IX and 10 show the seasons, working days, holidays and the probability of occurrence of the dates included in each type of curve for Data1 and Data2.

From Fig.12, it can be seen that the trends of these curves are basically the same from late night to early morning, and the load changes are in line with residents' daily routines. At the



Fig. 10. Cluster results of user load for data1.

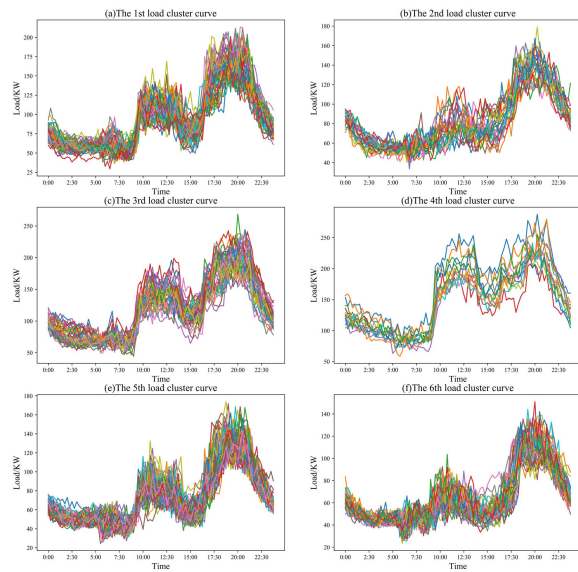


Fig. 11. Cluster results of user load for data2.

same time, the results in Table IX show that the 1st cluster and the 7th cluster belong to the typical winter load curve. Compared with other types of loads, the daily load peak-to-valley difference of these two types of curves is larger. The load value around 7:30 in the morning is higher than other types of curves. The time when electricity consumption starts to rise rapidly in the afternoon is basically concentrated at 3 o'clock. After 3 o'clock, the load has been increasing. Until about 21:00, the highest peak of daily load appears, and the load value is at a higher level in all curves. This is due to the

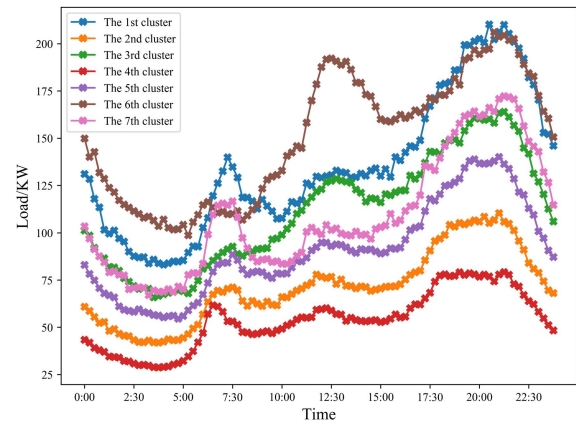


Fig. 12. Typical daily load curve for data1.

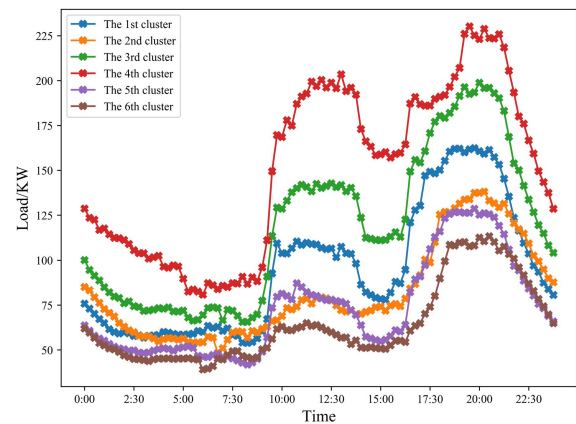


Fig. 13. Typical daily load curve autumn is concentrated between data2.

TABLE X
THE NUMBER OF DAYS INCLUDED IN EACH TYPE OF LOAD CURVE AND
THE PROBABILITY OF OCCURRENCE IN DIFFERENT
SEASONS FOR DATA2

Type of load curve	Number of days	Number of days /probability of occurrence		
		Spring&Fall	Summer	Winter
Cluster 1	54	8/0.148	12/0.222	34/0.629
Cluster 2	24	2/0.083	2/0.083	20/0.833
Cluster 3	39	0/0	17/0.436	22/0.564
Cluster 4	13	0/0	13/1	0/0
Cluster 5	77	68/0.883	7/0.091	2/0.026
Cluster 6	44	2/0.026	0/0	3/0.068

low temperature at night in winter, in addition to the use of air conditioners, heaters and other equipment for heating, the use of water heaters will also increase.

From the gray correlation analysis, we know that the main factors affecting load consumption in winter are relative humidity, daily rainfall, and average air pressure. By comparing the meteorological factor data composed of the 1st cluster and the 7th cluster load, we can find that the relative humidity in the 1st cluster load curve is generally higher than that of the 7th cluster, and the weather is mostly rainy, with high humidity

and cold weather. Therefore, the overall load value of the 1st cluster load curve is higher than that of the 7th cluster. The 6th cluster load curve is a typical summer curve. Through analysis, we find that the time for people to get up in the morning is earlier than in winter, and the load value reaches the first small peak at about 6:15. Due to the hot weather, air conditioning and other refrigeration equipment are used for a long time, which in turn keeps the summer load level at a high level. In summer, the loads represented in the 2nd, 3rd, 4th and 5th cluster curve contain more days, the curve trend is basically the same, and the load value gradually increases with time. The main difference between the four types of loads lies in the level of overall load consumption and the duration of the night load peak. The 4th cluster load curve contains the most days of spring & autumn and the lowest load level. This is because the air is cool in spring and autumn, the relative humidity is low, and the summer weather in the 4th cluster load curve is heavy rain, the temperature is relatively low. So the load value is small. In the 3rd cluster load curve, which contains the largest number of days in summer, the temperature difference causes the load values in the afternoon and night to increase significantly compared to the daytime.

From Fig.13, we can see that all load curves are bimodal curves, and reach the highest peak of daily load at night. At the same time, the results in Table X show that the 4th cluster load curve is a typical summer curve. Affected by high temperature, the overall load level of the 4th cluster load curve is significantly higher than that of the other clusters. Both peak loads exceed 200kw, and the peak-to-valley difference of daily load is large. The 3rd cluster load curve has the largest average daily precipitation among the all load curves, which leads to an increase in electricity load. This is because the rain in winter makes the weather wet and cold and reduces human comfort, which in turn causes users to use air-conditioning and other heating equipment to increase electricity consumption; while in summer, the high temperature and heavy rainfall make users need a clean and dry environment. The use of air-conditioning and other refrigeration equipment has greatly increased the electrical load. This is because the rain in winter makes the weather wet and cold and reduces human comfort, which in turn causes users to use air-conditioning and other heating equipment to increase electricity consumption; while in summer, the high temperature and heavy rainfall make users need a clean and dry environment. The use of air-conditioning and other refrigeration equipment has greatly increased the electrical load. The 6th cluster load curve is the type of curve with the lowest load level, which does not include the summer load type. From the weather data corresponding to the 6th cluster load, we know that the temperature in spring and autumn is concentrated between 8-18, and the date corresponding to this type of load curve has a smaller average relative humidity, which makes the human body more comfortable. As a result, fewer cooling and heating loads are used. Therefore, the overall load trend is stable and maintained at a low level. However, the 1st, 2nd and 5th cluster load curves include three seasons, and there are many

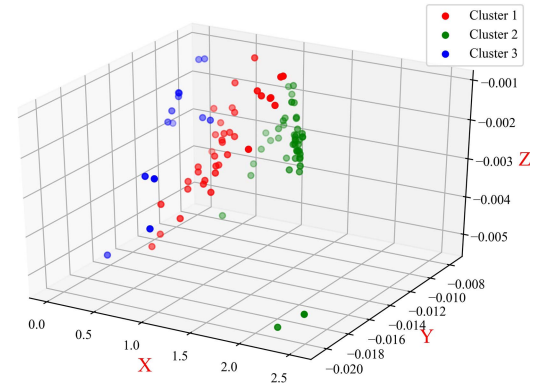


Fig. 14. Cluster results of electric vehicles in data3.

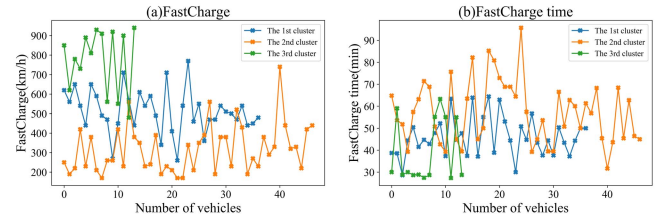


Fig. 15. Comparison of fastcharge and fastcharge time of three types of EVs after 1 hour of fast charging.

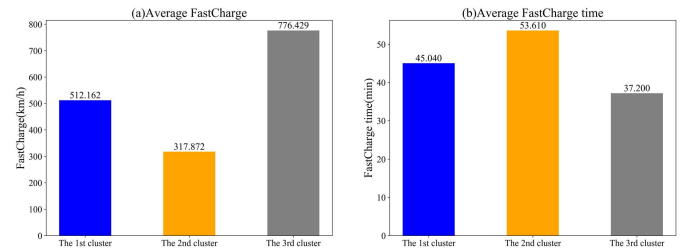


Fig. 16. Comparison of average fastcharge and average fastcharge time of three types of EVs after 1 hour of fast charging.

meteorological influence factors, which makes it impossible to accurately analyze the causes of user load changes.

D. Experimental Analysis of EV Charging Behavior

In this section, we perform standard discretization of the four features (PowerTrain, PlugType, BodyStyle and Segment) in Data3. Then we visualize the related features of electric vehicles based on AFS-SAEUL and the clustering labels based on UBA-AFSIC, and can intuitively find that the electric vehicles in Data3 are divided into three clustering, as shown in Fig.14. From Fig.14, we find that the cluster 1, 2 and 3 contain 37, 47, and 14 electric vehicles, respectively.

To understand the charging characteristics of electric vehicles, Fig.15 compares the FastCharge and FastCharge time of the three types of electric vehicles after one hour of fast charging. And, we also calculated the average FastCharge and average FastCharge time of the three types of electric vehicles after one hour of fast charging. The results are shown in Fig.16.

From Fig.15 and Fig.16, we can see that after all electric vehicles are charged for 1 hour, compared with electric vehicles in the 1st and 2nd clustering, the average travelable distance of electric vehicles in the 3rd clustering is the largest, reaching 776.429km. The average travelable distance of electric vehicles in the 1st and 2nd clustering is only 512.162km and 317.872km respectively. By calculating with the maximum mileage after fully charged, we also find that compared with electric vehicles in the 1st and 2nd clustering, the average charging time of electric vehicles in the 3rd clustering is the shortest, only 37.2 minutes. The average charging time of electric vehicles in the 1st and 2nd clustering is 45.04min and 53.61min, respectively. By analyzing other characteristic parameters of the three types of electric vehicles, we believe that the reasons are shown as follows.

(1) The driving method of electric vehicles makes the distances that can be traveled are different with the same battery capacity. All electric vehicles in the 3rd clustering are all-wheel drive; while some electric vehicles in 1st and 2nd clustering are front-wheel drive and some electric vehicles are rear-wheel drive.

(2) The type of charging plug of all electric vehicles may make the battery charging time different. The charging plugs of electric vehicles in the 3rd clustering are Type 2 except for Type 2 CCS; the charging plugs of electric vehicles in the 1st clustering are all Type 2 CCS, and the charging plugs of electric vehicles in the 2nd clustering are Type 2 CHAdeMO except for Type 2 CCS.

By analyzing the charging behavior of all electric vehicles in Data3, it can be judged that the overall performance of electric vehicles in the 3rd clustering is the best. However, from the experimental data, we also find that electric vehicle in the 3rd clustering has the highest price. This also provides a basis for users to purchase and maintain electric vehicles.

V. CONCLUSION

To better analyze user behavior under complex power grids, this article first proposes an adaptive feature selection algorithm of user behavior based on stacked autoencoder and unsupervised learning (AFS-SAEUL), selects typical user behavior characteristics, and reduces the complexity of user behavior data. Then we build a user behavior analysis model based on adaptive feature selection and improved clustering (UBA-AFSIC), using the improved K-Means clustering algorithm to achieve unsupervised learning of user behavior data. Experimental results on two real load datasets and a public electric vehicle dataset show that compared with other feature selection algorithms, the AFS-SAEUL algorithm proposed in this paper has a higher feature selection ratio, and the selected user behavior data has better unsupervised classification performance, and compared with user behavior analysis models based on other clustering algorithms, the UBA-AFSIC proposed in this paper has better clustering performance, that is, the lowest DBI and the highest SC. Meanwhile, based on the user behavior model obtained from the experiment, we know that the characteristics of electricity consumption are

related to the weather, and the charging time and mileage of electric vehicles are related to the charging plug and price. These conclusions provide theoretical basis for later user planning and marketing.

However, with the continuous increase of user behavior data and the extension of correlation analysis, the UBA-AFSIC model proposed in this paper still has the following problems: (1) the autoencoder parameters used in UBA-AFSIC are complex and have many combinations, and the model needs to optimize these parameters; (2) the efficiency of unsupervised learning is not discussed in this study. In addition, security and privacy are also the focus of user behavior analysis [47]–[49]. We will address these problems in the future.

APPENDIX A LIST OF ABBREVIATIONS

AFS-SAEUL	Adaptive Feature Selection algorithm of User Behavior based on Stacked Autoencoder and Unsupervised Learning
UBA-AFSIC	User Behavior Analysis model based on Adaptive Feature Selection and Improved Clustering
AE	AutoEncoder
SAE	Stacked AutoEncoder
DBI	Davies Bouldin Index
SC	Silhouette Coefficient
PCA	Principal Component Analysis
SRP	Sparse Random Projection
AC	Agglomerative Clustering
GMM	Gaussian Mixed Model
SC	Spectral Clustering
FCM	Fuzzy C-Mean
Mkmeans	Minibatch K-Means

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